

of scale, as these modules are mass-produced, resulting in significant cost savings compared to developing a custom, one-off solution.

Time-to-market. SoM accelerates time-to-market by employing pre-tested and certified modules. Instead of worrying about the complexity of core hardware, one can focus on application-specific requirements. Building custom hardware can be time-consuming, whereas SoMs accelerate development by providing a ready-made foundation, reducing the time required to bring your product to market.

Stability and reliability. SoMs are thoroughly tested and certified, ensuring hardware stability and reliability, reducing the risk of unforeseen issues in the development process. SoMs are designed to be scalable, allowing you to choose a module that suits your current needs and easily upgrade to a more powerful version as your requirements grow. They are often thoroughly tested and validated by the manufacturer, ensuring higher reliability and reducing the risk of hardware-related issues.

Focus on core competencies. By utilising an SoM, your development team can focus on the unique aspects of your project, such as software development and application-specific features, rather than getting bogged down in the intricacies of hardware design.

Opting for an SoM is a strategic choice that streamlines hardware development, accelerates progress, reduces expenses, and enhances adaptability. This decision empowers you to concentrate on crafting the optimal solution tailored to a specific industry or use case.

Selecting the right SoM

In the ever-evolving landscape of hardware development, one critical decision stands out as pivotal:

Simplifying high-speed electronics

Simplicity is key in high-speed electronics like 10Gbps or 5G. SoM design simplifies complex chip-down approaches, offering efficiency, cost-effectiveness, and fewer errors, especially in demanding applications.

Chip-down design complexity. Chip-down design custom-builds PCBs from scratch, integrating microprocessors, memory, and more for flexibility but adds complexity, especially in high-speed applications.

The SoM solution. It simplifies chip-down design challenges in high-speed applications. It is a compact, integrated computing platform with a single module housing a processor, memory, and more.

As the demand for high-speed electronics continues to grow, the advantages of SoMs over chip-down design become increasingly evident. SoMs simplify the development process, reduce complexity, and offer cost-effective solutions, particularly valuable in demanding applications. By harnessing the power of SoMs, engineers and developers can meet the challenges of high-speed electronics with confidence, efficiency, and reduced risk.

choosing the right SoM for your project. This crucial choice requires meticulous consideration of various factors to ensure the endeavor's success. The essential considerations that will guide you towards selecting the ideal SoM for your specific needs are as follows:

Form factor. Consider the physical space for hardware and any mechanical constraints. SoMs come in various forms, so choose one that fits your needs.

Cooling requirements. Determine whether the project requires a fanless system or if cooling mechanisms are necessary. Some processors generate more heat than others and may demand specific cooling solutions.

Interoperability. Ensure that the chosen SoM can support the desired software stack, including any operating systems or frameworks one plans to use.

Customisation. Evaluate whether the SoM manufacturer can accommodate customisation to match specific requirements.

Pin-to-pin compatibility. Check if the SoM offers pin-to-pin compatibility for easy hardware upgrades or changes.

Customer support. The availability of reliable customer support, comprehensive documentation,

and a helpful community can be invaluable during the development journey.

Certifications. Depending on the application, certifications like EMI/EMC, safety standards, and compatibility with industry-specific regulations may be essential. Ensure the chosen SoM aligns with these requirements.

When it comes to hardware development, SoMs offer a bridge to the future where modularity and adaptability reign supreme, ensuring that your hardware solutions remain agile and competitive in an ever-changing technological landscape. It is time to leverage SoMs as a powerful tool in the hardware development toolkit and confidently embrace the future. SoMs have emerged as a powerful solution in the world of hardware development. Whether working on IoT projects, automotive applications, or any other hardware endeavor, SoMs provide a versatile foundation to build upon. As technology evolves, SoMs will play an increasingly pivotal role in the hardware landscape, simplifying development and accelerating innovation. **EFY**

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